Market Technicians Association, Inc.
Professionals Managing Market Risk • Incorporated in 1973
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Chartered Market Technician (CMT) Program – Level I

The CMT Level I exam measures basic competence of an entry-level analyst. The CMT Level I candidate should have a working knowledge of the terminology used in the required readings, be able to **identify** the concepts discussed in these readings, and have a definitional understanding of the analytical tools presented in the required readings.

Exam time length: 2 hours, 15 minutes
Exam format: Multiple Choice

The curriculum is organized into exam specific knowledge domains that provide a framework for recognizing and implementing investment/trading decisions. CMT Level I exam tests the candidate's knowledge in 12 domains:

1. Theory and History
2. Markets
3. Market Indicators
4. Construction
5. Trend Analysis
6. Chart and Pattern Analysis
7. Confirmation
8. Cycles
9. Selection and Decision
10. System Testing
11. Statistical Analysis
12. Ethics

CMT®
CHARTERED MARKET TECHNICIAN

CMT Level I Exam Topics & Question Weightings

1. Theory and History	a.	history of financial markets (e.g., historical market events, bubbles, crashes)	9%	11
	b.	history of technical analysis		
	c.	modern portfolio theory (e.g., efficient market hypothesis, random walk)		
	d.	Dow theory		
	e.	behavioral finance		
2. Markets	a.	historical market data (e.g., non-trading days/hours, characteristics of various markets)	5%	6
	b.	traditional asset classes (e.g., equities, fixed income, commodities)		
	c.	alternative asset classes (e.g., derivatives, private equity, managed futures, real estate)		
	d.	currencies		
	e.	non-US markets		
	f.	market indices		
	g.	exchanges		
3. Market Indicators	a.	breadth indicators (e.g., A/D, up/down volume)	7%	9
	b.	index construction		
	c.	government/Fed reports (e.g., treasury data, monetary policy, Fed holdings, CPI)		
	d.	private money flows (e.g., mutual fund holdings, corporate liquidity holdings)		
	e.	sentiment measures (e.g., put-call ratio, investor polls)		
	f.	volatility (e.g., vix, historical, implied)		
4. Construction	a.	scaling methods (e.g., arithmetic, semi-log)	5%	6
	b.	line chart		
	c.	bar chart		
	d.	candlestick chart		
	e.	point and figure chart		
	f.	volume		
5. Trend Analysis	a.	trendlines	16%	18
	b.	regression analysis		
	c.	moving averages		
6. Chart and Pattern Analysis	a.	classic pattern recognition	23%	28
	b.	candlestick pattern recognition		
	c.	Elliott Wave principle		
	d.	Fibonacci price analysis		

	e.	support and resistance		
	f.	relative strength index (RSI)		
	g.	moving average convergence/divergence (MACD)		
	h.	Bollinger Bands		
	i.	stochastics		
7. Confirmation	a.	open interest and volume	3%	4
8. Cycles	a.	business cycles	5%	6
9. Selection and Decision	a.	relative strength	13%	15
	b.	forecasting techniques (pattern and trend recognition)		
10. System Testing	a.	backtesting	5%	6
	b.	order execution and slippage		
11. Statistical Analysis	a.	descriptive statistics (e.g., mean, median, mode,	6%	7
	b.	fundamentals of probability		
12. Ethics		Ethics Standards and Practices	3%	4

CMT Level I Exam - Learning Objectives

1. Theory and History	
history of financial markets (e.g., historical market events, bubbles, crashes)	Describe the development of modern technical analysis. Describe historically normal price action in both rising and falling markets. Compare the market behavior considered "normal" and various historical events where price action moved in unexpected ways.
history of technical analysis	Describe the origins of technical analysis. List the underlying assumptions of technical analysis. Describe the essence of the study of technical analysis. Contrast technical analysis and fundamental analysis. List the characteristics of markets in which technical analysis is most effectively used.
Modern Portfolio Theory (e.g., Efficient Market Hypothesis, Random Walk)	Discuss the pragmatic criticisms of technical analysis. Discuss the assumptions of technical analysis. Identify the basic concept of the Efficient Market Hypothesis (EMH). Describe how technical analysis remains relevant despite the EMH.
Dow theory	Describe the history of the original Dow indexes. Identify the basic principles of Dow Theory.
behavioral finance	Describe the concept of fungibility. Recognize the characteristics of stock prices as a martingale. Identify valid challenges to EMH. Identify valid criticisms of the CAPM model. Identify the definition of a Noise Trader. Explain why Technical Traders are considered a specific type of noise trader. Explain the implications of Technical Traders in the market.
2. Markets	
historical market data (e.g., characteristics of various markets)	Describe how market prices are quoted and how they change in various markets.
traditional asset classes (e.g., equities, fixed income, commodities)	Describe the differences between stocks and bonds. Identify reasons why an investor might prefer stocks, bonds or commodities. Describe the implications of widening or narrowing of yield spreads on bond prices. Describe the normal relationship between yield spreads and bond ratings .

alternative asset classes (e.g., derivatives, private equity, managed futures, real estate)	Discuss leverage in the context of futures markets versus cash markets.
currencies	Recognize the differences between the forex market and other markets. Recognize relative values between various currencies.
non-US markets	Recognize how stocks are quoted in Non-US markets. Recognize impact of US markets on non-US markets.
market indices	Identify most commonly used market indices.
exchanges	Identify most commonly used exchanges. Recognize which markets are served by various exchanges.
3. Market Indicators	
breadth indicators (e.g., A/D, up/down volume)	Describe the concept of breadth. Explain what changes in breadth may signal about the markets. List the various breadth indicators. Describe new highs and new lows as a breadth indicator. Describe the advance-decline line and how it is used.
index construction	Identify the different weightings that may be used in an index. Identify the specific weightings used for commonly followed indexes. Explain how stock price changes impact price weighted, market capitalization-weighted and equally weighted indexes.
government/Fed reports (e.g., Treasury data, monetary policy, Fed holdings, CPI)	Describe the yield curve. Interpret what changes in the yield curve signals about future market performance. Describe the main tools that the Federal Reserve uses to adjust the money supply. Interpret the likely results of changes in Fed monetary policy tools on market conditions.
private money flows (e.g., mutual fund holdings, corporate liquidity holdings)	Describe mutual fund cash as a percentage of total assets as both a sentiment indicator and a flow of funds indicator. Describe insider trading as a sentiment indicator.
sentiment measures (e.g., put-call ratio, investor polls)	Describe various sentiment indicators. Describe the put/call ratio. Interpret the signals provided by the put/call ratio. Describe the use of advisory service opinion as a sentiment indicator. Describe the logic behind using margin debt as a sentiment indicator.
volatility (e.g., vix, historical, implied)	Describe the difference between historical and implied volatility Describe the VIX index. Explain the implications of a rising or falling VIX index.

4. Construction	
scaling methods (e.g., arithmetic, semi-log)	Explain the arithmetic and semi-logarithmic scaling conventions. Contrast the advantages and disadvantages of arithmetic and semi-logarithmic scales in charting.
line chart	Describe a line chart. Compare and contrast line charts with other chart types.
bar chart	Describe a bar chart. Compare and contrast bar charts with other chart types.
candlestick chart	Compare and contrast candlestick charts with other chart types. Describe the construction of a candlestick chart. Describe the advantages of a candlestick chart. Define real body and shadow/wick. Describe how candlestick analysis can be used in conjunction with Western techniques.
point and figure chart	Describe how point and figure charts are constructed. Describe the importance of box size on the sensitivity of point and figure charts. Construct various box size reversal point and figure charts. Interpret reversal signals on a point and figure chart. Describe the concept of price targets attained by using a horizontal or vertical count on a point and figure chart.
volume	Describe the importance of volume data in technical analysis. List the general rules regarding changes in volume and signals of market activity. Interpret price and volume charts. Describe the role volume and volume spikes play in determining turning points in the market. Describe volume patterns in typical technical formations (head and shoulders, wedges, triangles, flags, pennants, etc.). Describe the concept of on-balance volume. Interpret a chart containing price and on-balance volume.
5. Trend Analysis	
trend lines	Describe the characteristics of trends. Discuss the uses of trend lines. Discuss the factors in determining the significance of a trendline
regression analysis	Describe what slope indicates in a trendline regression equation. Describe the use of regression line studies to determine trends in price data.
moving averages	Describe a moving average. Explain the logic of using moving averages as trend indicators. Describe the use of different length moving averages in a trading system.

	Contrast simple and exponentially weighted moving averages. Describe the buy and sell signals resulting from moving average crossovers.
6. Chart and Pattern Analysis	
classic pattern recognition	Identify variations price pattern formations (including but not limited to: triangles, flags, pennants, wedges, double and triple tops, double and triple bottoms). Describe head and shoulders and inverse head and shoulders formations. Describe research associated with head and shoulders patterns. Describe the typical volume relationships in price pattern formations. Identify and discuss the implications of broadening patterns. Identify and discuss the implications of rounding formations.
candlestick pattern recognition	Identify various single candle formations including primarily, though not limited to, the following: maribozu, spinning tops, dojis, hammers, shooting stars, high wave candles, and windows and their implications. Identify multi-candle reversal patterns such as engulfing candles. Discuss the implications of various candle formations.
Elliott Wave principle	Describe the important elements of the Elliott Wave Theory. Differentiate between impulse waves and corrective waves. Identify the count of a given wave in a defined structure.
Fibonacci price analysis	Describe the Fibonacci sequence. Apply Fibonacci ratios to price and time targets. List the Fibonacci ratios used in retracements.
support and resistance	Describe the concepts of resistance and support. Define acceptable levels of penetration to confirm a break of support or resistance. Explain the concepts of a trading range and support and resistance levels.
relative strength index (RSI)	List the uses for an oscillator. Describe how the choice of time period influences the signals provided by oscillators. Identify main components of the RSI. Interpret RSI readings for overbought and oversold levels.
moving average convergence/divergence (MACD)	Describe the main components of the MACD. Identify potential trading signals using a MACD indicator.
bollinger bands	Describe the components of Bollinger Bands compared with moving average envelopes. Discuss the significance of Bollinger Bands widening or narrowing.

stochastics	Describe how stochastic oscillators are computed. Differentiate between a fast and slow stochastic. Interpret changes in stochastic indicators including crossovers, divergence failure, reverse divergence, extremes, hinges, and divergences.
7. Confirmation	
open interest and volume	Define the concept of open interest. Contrast open interest and daily volume. Describe volume patterns in typical technical formations (head and shoulders, wedges, triangles, flags, pennants, etc.). Describe the concept of on-balance volume. Interpret a chart containing price and on-balance volume.
8. Cycles	
business cycles	Describe market cycles and how they differ from mathematical cycles. Describe the basic principles of cycles. Describe seasonal patterns in the markets.
9. Selection and Decision	
relative strength	Define relative strength (RS). Describe how relative strength (RS) is used. Describe the value of relative strength study. Identify a correlation coefficient. Describe how a correlation coefficient is calculated. Interpret the meaning of a given correlation coefficient.
forecasting techniques (pattern and trend recognition)	Discuss the basic premise of intermarket analysis. Describe the concept of sector rotation. Identify leading and lagging industry groups.
strategic models	Identify the basic components of a forecasting model. Identify the basic components of an investing strategy.
10. System Testing	
backtesting	Discuss the parameters of an appropriate system of backtesting. Describe how to interpret the typical results of backtests.
order execution and slippage	Explain the impact of order execution and slippage on the cost basis of an investment. Explain the impact of slippage on backtest results.
11. Statistical Analysis	
descriptive statistics (e.g., mean, median, mode)	Explain the differences between traditional backtesting technique and quantitative, or statistical, analysis. Distinguish between the arithmetic mean and geometric mean. Describe how the relationship between variables is determined.

fundamentals of probability	Describe the parameters of a normal distribution. Describe how the actual distribution of stock returns deviates from the normal distribution. Define kurtosis.
12. Ethics	Code of Ethics and Standards of Professional Conduct